

Spanish Maritime Empire: Capability Tier Analysis (1492-1648)

Session 2 of 4: Complex Adaptive Systems Framework

Spain's Capability Tier reveals a fundamental paradox central to the 1694 thesis: the empire possessed substantial social mobility channels, world-leading educational institutions, and Europe's greatest cultural production during economic decline, yet these capabilities failed to translate into financial innovation or sustained hegemonic advantage. This session documents how Spain scored higher than Portugal across all five capability domains while demonstrating why resource abundance without institutional adaptation produces stagnation rather than transformation.

Session 2 executive summary with preliminary scores

The Capability Tier analysis confirms Spain's structural advantages over Portugal while revealing systematic barriers to converting those advantages into lasting power. **Key findings:** The anti-commercial "hidalgo mentality" was significantly overstated by traditional historiography—regional variation was substantial, with Basque and Catalan commercial cultures thriving alongside Castilian disdain for trade. *Limpieza de sangre* enforcement was negotiable through purchased genealogies and bribed witnesses, creating a system that was "rhetorically rigid but operationally porous." Most striking is the **Golden Age paradox**: Spain produced Europe's greatest literary and artistic output precisely during its steepest economic decline, explained by José Antonio Maravall's "cultura dirigida" thesis—Baroque culture as conservative response to crisis rather than despite it.

Preliminary Domain Scores (Portuguese baseline in parentheses):

Domain	Rise 1469-1556	Peak 1556-1598	Decline 1598-1648
Social/Class	2.5 (2.5)	2.5 (2.5)	2.0 (2.0)
Educational	3.75 (2.0)	4.25 (2.0)	3.25 (1.5)
Cultural	3.5 (3.0)	4.5 (2.5)	5.0 (2.0)
Technological	3.75 (3.5)	4.25 (3.0)	2.75 (2.0)
Demographic	3.25 (2.5)	2.75 (2.0)	1.75 (1.5)

Average Spanish Capability Tier: Rise 3.35, Peak 3.65, Decline 2.95 **Average Portuguese Capability Tier:** Rise 2.70, Peak 2.40, Decline 1.80 **Spanish Advantage:** +0.65 (Rise), +1.25 (Peak), +1.15 (Decline)

Social and class structure analysis

The hidalgo myth unravels under regional scrutiny

Traditional historiography portrayed Spain as uniformly anti-commercial, with the hidalgo mentality representing a society-wide disdain for productive labor. Ruth MacKay's revisionist scholarship in "*Lazy, Improvident People*" (2006) demolishes this narrative, demonstrating it was largely a **myth created by eighteenth-century Enlightenment intellectuals** who disparaged earlier generations to advance their own reform agenda. (OverDrive) MacKay's archival research reveals that "early modern artisans...were neither particularly slothful nor untalented. On the contrary, they behaved as citizens, and their work was seen as dignified and essential to the common good." (OverDrive)

Regional variation was stark. **Castile** represented the heartland of hidalgo culture, where disdain for "mechanical and commercial arts" (Duke University Press) concentrated most heavily. Yet even here, the picture was complex—the letrado class of university-trained bureaucrats provided an alternative path to status through professional competence rather than hereditary nobility. **The Basque Country** presented a dramatic contrast: Basques "thrived on whale hunting, shipbuilding, iron exportation to England, and trade with northern Europe" throughout the sixteenth century. (Wikipedia) (Wikipedia) Basque iron production exceeded **3,000 tons annually**, (Uca) and near-universal hidalguía paradoxically enabled commerce—in Santander's Catastro of Ensenada, "almost all registered residents were nobles, despite being peasants or working as masons, blacksmiths and other trades." (Wikipedia) The Basques formed powerful merchant diasporas, comprising 48% of immigrants to Chile by the seventeenth century. (Wikipedia)

Seville represented an exceptional urban commercial culture documented by Ruth Pike. The city's position as Indies trade monopoly hub created conditions where "aristocrats...rolled up their gilt-edged sleeves and plunged into wholesale financing of commercial ventures." Merchants purchased noble titles while continuing trade, (Duke University Press) and even local nobility "engaged in trade in the city"—a stark departure from aristocratic norms. (Tontinecoffeehouse) Seville grew from **49,395 inhabitants in 1530 to 150,000 by the late sixteenth century**, (Spain Then and Now) becoming Spain's largest city and third-largest in Western Europe.

Limpieza de sangre was negotiable through corruption and forgery

The *limpieza de sangre* (blood purity) system presents a case study in the gap between formal requirements and operational reality. Formal statutes proliferated from the inaugural Sentencia-Estatuto of Toledo (1449) (Grokikipedia) through mid-century, with proof requirements extending to government offices, military orders, ecclesiastical positions, universities, guilds, and religious confraternities by the 1550s. The statutes remained in force until **1865**. (Wikipedia)

Yet enforcement was systematically undermined through documented evasion practices. In Baza, "judeoconverso-descended families systematically manipulated family trees...involving alterations to baptismal certificates and fabricated kinship ties." (Grokikipedia) One Toledo witness testified that an individual "gave mil ducados (1,000 ducats) in money to a certain person [the corregidor]...so that he could prove limpieza while not having a drop of pure blood." Henry Kamen's revisionist assessment concludes that "limpieza de sangre ultimately became a qualification negotiated through bribed witnesses, shuffled genealogies, and falsified documents." (Encyclopedia.com) Even the Duke of Olivares, Philip IV's chief minister, opposed the system.

María Elena Martínez's Stanford study *Genealogical Fictions* (2008) demonstrates how *limpieza* concepts transplanted to the Americas adapted to colonial conditions. The colonial caste system (*sistema de castas*) was similarly porous. Pilar Gonzalbo's 2013 study *La trampa de las castas* "discards the idea of a 'caste system' or 'caste society' in New Spain, understood as 'social organization based on race and supported by coercive power.'" [Wikipedia](#) Parish records show only five main terms actually used (Spanish, Indian, mulatto, mestizo) rather than the elaborate sixteen to twenty-two categories depicted in casta paintings. [Wooster](#) Mobility pathways included "blood mending" through documentation changes, whitening progression across generations, and purchased classifications—since the late sixteenth century, mestizos could "purchase from the Crown a certificate classifying them as 'Spaniards.'" [Wikipedia](#)

Social mobility channels remained operative despite formal restrictions

Four primary channels facilitated social advancement. The **letrado bureaucracy** provided approximately **400-500 positions** appointed by the Crown by the early seventeenth century, requiring university education but enabling non-noble advancement through professional competence. Richard Kagan documents how letrados "performed functions so indispensable to society and the state that they could not be denied social honor and status." [Duke University Press](#)

Church careers represented a more open estate where individuals could join without regard to social status, though hierarchy existed within Church structures. [Wikipedia](#) **Military service** through military orders (Santiago, Calatrava, Alcántara, Montesa) required *limpieza* proofs that could be purchased or forged. Most dramatically, **Indies trade** offered rapid mobility—Pike documents how merchants in Seville "utilized a portion of their profits to purchase noble titles and administrative posts," [Duke University Press](#) while the widespread use of *mayorazgo* (entailed estates) "reflects the city's dramatic social mobility" and "indicates the rapid ennoblement of Sevillian society." [Oxford Bibliographies](#)

Scoring Rationale: Spain's social/class scores match Portugal's (Rise 2.5, Peak 2.5, Decline 2.0) because despite greater institutional complexity and regional variation, both empires exhibited similar constraints: rigid formal hierarchies undermined by corruption, religious discrimination reducing commercial capacity, and declining mobility in late periods. Spain's larger scale created more total mobility opportunities but similar structural barriers.

Educational and knowledge systems analysis

Spanish universities created Europe's best-educated bureaucratic class

The Spanish university system represented a critical capability advantage over Portugal, which possessed only Coimbra as a major university. Spain operated multiple institutions of international significance. **Salamanca** enrollment data from matriculation records shows dramatic growth: **439 students in 1335**, rising to **6,328 by 1552**, and peaking at **6,778 in 1584**—"nearly 7,000" confirmed across multiple sources. This peak made Salamanca comparable to Oxford and Paris. Decline followed: **4,000 students by 1601**, **3,908 by 1641**, and only **2,000 by 1701**.

Salamanca's intellectual output was substantial. Francisco de Vitoria's lectures (1526-1546) established fundamental principles of international law. [\(Philopedia\)](#) The Salamanca School produced major theological, legal, and economic scholarship. Richard Kagan's thesis is emphatic: by 1600, Castile was "perhaps the best-educated kingdom in Europe." [\(Project MUSE\)](#) [\(Johns Hopkins University Press\)](#) The close connection between universities and bureaucratic institutions determined enrollment patterns — [\(Johns Hopkins University Press\)](#) the *letrado* class staffing royal councils emerged from these institutions.

Alcalá de Henares (founded 1499 by Cardinal Cisneros) represented a revolutionary humanist institution, the "world's first planned university city." [\(UAH\)](#) [\(Grokopedia\)](#) Its Trilingual College (*Patio Trilingüe*) taught Latin, Greek, and Hebrew, [\(Viajeraymochilero\)](#) while the **Complutensian Polyglot Bible** (1502-1517)—the first multilingual printed edition of the full Bible—represented a major scholarly achievement. Notable alumni included Lope de Vega, Francisco de Quevedo, and Ignatius of Loyola.

Valladolid served as the principal legal training center for Castile, training *letrados* for *chancillería* courts. Kagan identifies graduates dominating crucial positions in royal councils including Castile, Cámara, and Military Orders. [\(Duke University Press\)](#)

The Jesuit network created unprecedented global knowledge transfer

The Society of Jesus built history's first truly global educational network. By **1556** (Ignatius's death): 1,000 Jesuits operating 35 schools. By **1581**: 144 schools. By **1616**: nearly 400 schools. By **1700**: 700+ colleges and universities in Europe alone. By **1749** (pre-suppression): 800+ institutions worldwide.

Scientific contributions were remarkable. Matteo Ricci in China (1582-1610) introduced European astronomy, mathematics, and cartography to the imperial court, translating Euclid's *Elements* into Chinese (1607)—described as the "crown of Western studies." Johann Adam Schall von Bell developed a new calendar for the Qing dynasty and was appointed Director of the Imperial Astronomical Bureau (1645). **Over 130 European scientific titles were translated into Chinese** in astronomy, mathematics, geography, medicine, and physics. The network also produced the first systematic grammars and dictionaries of numerous colonial languages.

Colonial universities expanded European institutional models but restricted access

Spain established the Western Hemisphere's first universities. **Santo Domingo** (1538) represented the inaugural institution. **Lima (San Marcos, 1551)** and **Mexico City (1551)** followed, both founded by royal decree in the same year. San Marcos operated five faculties (Arts, Theology, Medicine, Civil Law, Canon Law) with approximately **35 academic chairs** at its peak, modeled on Salamanca.

However, access was severely restricted. Universities "primarily served the upper classes" with "indigenous peoples and mestizos largely excluded from higher education." Access remained "largely confined to peninsular Spaniards and creoles." Alternative indigenous education through Franciscan institutions like Colegio de Santa Cruz de Tlatelolco (1536) aimed to train a native priesthood but was deemed a "failure" in that goal, though it produced a small cohort of trilingual indigenous intellectuals.

Inquisition suppression was real but more limited than the Black Legend suggests

The critical question—whether Inquisition censorship actually suppressed scientific and commercial knowledge—requires nuanced assessment. Henry Kamen's revisionist scholarship argues that "practical limitations ensured inquisitorial censorship was in general never very effective" and that the Inquisition was "institutionally flawed, only sporadically effective both temporally and geographically, and not ideologically monolithic."

Evidence supports limited direct scientific suppression. The "Spanish Inquisition never persecuted scientists" and a "relatively small number of scientific books were placed on Index." (Art and Popular Culture) Primary targets were conversos, Moriscos, Protestant literature, and illuminist spirituality—not scientists. Irving Leonard demonstrated that prohibited books like romances of chivalry reached the New World "with blessing of the Inquisition." (LiquiSearch) Many noblemen's libraries contained banned books.

Yet indirect effects were significant. Recent research by Cox and Figueroa (2025) emphasizes **self-censorship** and **reduced scholarly interactions** as more important than direct targeting. The imprisonment of Luis de León (1572-1576) for translating the Song of Songs from Hebrew demonstrates suppression of textual and philological work. The Index of 1559 was "the only repressive Index prior to the eighteenth century" but its severity created lasting chilling effects. José Pardo Tomás's *Ciencia y Censura* documents that while science was not a primary target, secondary effects on scientific culture occurred.

Scoring Rationale: Spain scores significantly higher than Portugal on education (Rise 3.75 vs. 2.0, Peak 4.25 vs. 2.0, Decline 3.25 vs. 1.5) due to multiple major universities versus Portugal's single Coimbra, the global Jesuit network, colonial university establishment, and Casa de Contratación's cartographic enterprise. The decline score remains high because despite enrollment drops, institutional infrastructure persisted.

Cultural and ideological systems analysis

Counter-Reformation patronage created distinctive Spanish visual culture

The Council of Trent (1545-1563) profoundly shaped Spanish cultural production. (Philopedia) Religious art patronage concentrated in ecclesiastical institutions rather than courts. **El Greco** (1541-1614) combined Byzantine traditions with Venetian technique and Counter-Reformation spirituality— (Substack) his elongated figures and mystical intensity represented "the most vital visual representative of Spanish mysticism."

(Encyclopedia Britannica) Rejected by Philip II for El Escorial, he found ecclesiastical patronage in Toledo.

(Wikipedia)

Francisco de Zurbarán (1598-1664), described as "after Velázquez, the greatest painter of the Golden Age," worked primarily for monastic patrons in southern Spain, not the court. (Metropolitan Museum of Art) His austere, tenebrist style perfectly expressed Counter-Reformation theology. The Church used art explicitly as Counter-Reformation instrument to "communicate religious themes" and inspire devotion— (Aesthetics Wiki) Zurbarán's invitation from Seville's city government stated that art was "one of the major embellishments of the state."

(Metropolitan Museum of Art)

Religious drama through the *auto sacramental* genre reached its apex. These one-act allegorical plays focused on the mystery of the Eucharist, performed during Corpus Christi on mobile stages throughout Spanish cities. Characters personified abstract concepts: Faith, Hope, Sin, Death, Grace, Nature. Calderón described his autos as "sermons set in verse, problems of Sacred Theology set in representable ideas." The genre peaked in the seventeenth century before being banned in 1765 by Carlos III.

The Golden Age paradox demands explanation

The central puzzle of Spanish cultural history: why did Spain produce Europe's greatest literary and artistic output during economic decline? Economic indicators show Spanish GDP per capita peaked in the 1590s then declined to all-time low by 1640. (Academia.edu) Philip II declared his third bankruptcy in 1596. (StudySmarter +2) Silver flood from Potosí hollowed out manufacturing through "Dutch Disease" — Santander's wool exports collapsed from 17,000 sacks on 66 ships (1572) to 505 bags on 11 ships (1622). (Berkeley Economic Review)

Yet cultural output flourished. Cervantes published *Don Quixote* (1605/1615) during this crisis. (Newberry Library) Velázquez's greatest works emerged from the 1620s-1650s. Lope de Vega wrote **1,500-1,800 plays** (400+ surviving). (En Academic) Calderón's masterpieces appeared 1629-1681. The paradox is real and demands explanation.

José Antonio Maravall's "cultura dirigida" thesis provides the most compelling framework. Baroque culture was a *response* to crisis, not despite it. Ruling classes mobilized mass culture to ensure "popular identification with sovereign power." Theatre and art served as "techniques of mass social operation" for conservative goals. The Baroque was "efficacious" in achieving "social and political effects of a conservative character." Critics charge Maravall presents too "univocal a paradigm," but the crisis-response framework illuminates the paradox.

Royal and aristocratic patronage networks sustained production. Philip IV (r. 1621-1665) was the most significant artistic patron despite political weakness. Velázquez served as court painter from 1623 until his death in 1660. The relationship was "deeply symbiotic": Velázquez projected Philip's image while Philip elevated Velázquez to nobility. Philip appointed Velázquez Palace Marshal (1658), combining administrative and artistic roles. (Wikipedia) The Buen Retiro Palace (1633) became a venue for elaborate court theatrical productions.

Urban commercial theatre created independent revenue streams. *Corrales de comedias* (permanent open-air theaters) built in major cities were originally licensed to charitable organizations supporting hospitals. This created a thriving commercial theatre industry with professional troupes, diverse audiences from standing patio to box seats, and accessibility to all social classes. Pierre Vilar's analysis connects *Don Quixote* directly to economic crisis: it "only gathers its full meaning in its true historical setting," confronting obsolete chivalric ideals against emerging capitalist morality. (New Left Review)

Regional variation in commercial attitudes undermines monolithic narratives

Evidence directly challenges the "all Spain was anti-commercial" narrative. **Catalonia** maintained Mediterranean commerce traditions through the Crown of Aragon heritage, with the Consulate of the Sea producing pioneering maritime and commercial law documents that ruled Mediterranean transactions for centuries. Trade networks extended to Occitania, France, Italy, the Maghreb, and the Mediterranean Levant.

However, marginalization under Castilian dominance included the 1518 prohibition on Catalan fleet trading with the Americas.

Basque Country merchants thrived as "masters not only of whaling but the Atlantic Ocean" by the sixteenth century. (Wikipedia) Magellan's expedition included at least 35 Basque sailors, and Juan Sebastián Elcano completed the first circumnavigation. (Wikipedia) The Royal Company of Caracas (1728) demonstrated continued Basque commercial capacity trading iron, wool, and wines for colonial products. (ResearchGate) Customs privileges through *fueros* maintained free markets for imports and exports.

Honor culture constrained economic behavior but less than traditionally claimed

Honor (*honra*) was "a major theme in Spanish Golden Age drama"—Calderón's *El médico de su honra* represents the "most telling example" where plot revolves around announced death of an innocent woman suspected of adultery. Maravall identified honor as "one of the principal axes of Spanish society."

(Oxford Bibliographies) *Limpieza de sangre* documents required proving ancestry was "old Christian," free of Inquisition accusations, without associations with "manual or dirty occupations." (Oxford Bibliographies)

(Sephardichorizons)

Economic implications were real but mediated by regional variation. Manual labor and commerce associated with "tainted" blood deterred commercial participation by those seeking social mobility. Yet the documented commercial cultures of Seville, the Basque Country, and Catalonia demonstrate that honor constraints were unevenly enforced.

Scoring Rationale: Cultural scores show Spain dramatically outperforming Portugal, especially in the Decline phase (5.0 vs. 2.0). The paradoxical Golden Age flowering during crisis represents unique adaptive capacity—culture responded to systemic stress through intensification rather than collapse. Portugal's single great literary figure (Camões) contrasts with Spain's broad cultural production across literature, visual arts, and drama. The Decline score of 5.0 reflects the counterintuitive reality that Spain's cultural PEAK came during economic DECLINE.

Technological and innovation capacity analysis

Mining technology represented Spain's most significant original contribution

The **patio process** (mercury amalgamation) invented by Bartolomé de Medina at Pachuca, Mexico in 1554–1557 (HISTORY) was revolutionary—enabling silver extraction from low-grade ores with **70-80% recovery rates**. The technique remained in use until 1870. (Duke University Press) The *magistral addition* (copper sulfate catalyst) dramatically improved efficiency. Pan amalgamation (1609) at Potosí adapted the process to Andean conditions, (Wikipedia) reducing processing time from 6-8 weeks to 10-14 days.

The hydraulic infrastructure at Potosí represented large-scale industrial engineering: **22 artificial lakes** (*lagunas*) and **140 water-powered mills** (*ingenios*) for ore grinding, created under Viceroy Toledo's reforms in the 1570s. Silver production data demonstrates scale: **60,000 tons** total from Potosí (1545–1810), (Wikipedia)

with peak 1590s-1600 representing **60% of world silver production**. Annual output reached **2,700 tons** circa 1600. (Money Museum) Spanish America produced **136,000 metric tons** total (1550-1800), representing 80% of global supply. (Wikipedia)

Mercury from Huancavelica (Peru's "mine of death") sustained this production—approximately **36,000 tons** extracted (Earth Magazine) (1564-1810). The mercury-silver ratio of 2:1 (two pounds mercury per pound silver) created a critical supply chain dependency.

Navigation achievements opened the Pacific but did not sustain technological leadership

Andrés de Urdaneta's discovery of the *tornaviaje* (return route) in 1565 via the Kuroshio Current at 38-40°N latitude represented a major navigational achievement, enabling the **129-day eastward crossing** that made the Manila Galleon trade viable. The Pacific route spanning **11,500+ miles** was standardized between 1565-1620, operating 108 ships over 250 years (1565-1815) with only 26 losses to enemy action or weather.

The **Casa de Contratación** maintained the Padrón Real (master map) from 1508, "constantly improved" throughout the sixteenth century and serving as "template for maps carried by every Spanish ship." Juan López de Velasco's cartographic enterprise "surpassed anything done by other European powers." Yet by century's end, Spanish intellectuals lamented that "maps produced by foreigners were superior to those made in Spain."

(Infogalactic)

Military innovation through the tercio eventually became obsolete

The tercio formation, developed by Gonzalo Fernández de Córdoba (the "Gran Capitán") from the 1490s-1530s, combined pike and shot integration in units of theoretically 3,000 men (practically 1,000-2,000 after campaigns). Pike-to-shot ratios started at 1:1 but evolved to heavily favor shot by the seventeenth century. Professional volunteers rather than conscripts or mercenaries provided superior quality.

The **Citadel of Jaca** (1590s) represented Spain's adoption of *trace italienne* fortification for cannon defense.

(Cabinet) However, Spanish military adaptation stalled. The Battle of Rocroi (1643) marked the "definitive end of Spanish military hegemony"—the tercio system was abandoned for French line infantry doctrine.

Why no sustained technological lead despite Potosí resources?

Multiple factors explain Spain's failure to convert resource abundance into sustained technological advantage. **Institutional factors** predominated: Counter-Reformation intellectual isolation led to "a general stagnation of the institution" as universities renewed under humanism fell into isolation from outside influences. Courtly patronage corruption meant "the dynamics of rivalry, patronage, interest groups and clientelism could be more important than the actual quality of the project or invention." (Springer) Patent systems functioned similarly to *arbitrios* (policy proposals), with success depending more on patronage networks than innovation merit.

Economic factors compounded institutional barriers. The "silver curse" caused **400% inflation** over the sixteenth century, making Spanish exports uncompetitive as wages rose. Serial bankruptcies (1557, 1560, 1575, 1596, 1607, 1627, 1647) drained capital from productive investment. (Lumen Learning) (Lumen Learning) Mercury dependency created supply chain vulnerabilities.

Technological stagnation evidence is clear. Mining: "No significant changes occurred" in amalgamation after 1600—71% of Mexican silver was still produced by the 1554 technique as late as 1870. Shipbuilding: By the seventeenth century, Dutch fluyt ships carried twice the cargo at half the cost with smaller crews. The Dutch advantages were structural: cheap energy (windmills, peat transported by canal), financial innovation (Amsterdam Stock Exchange 1602, Bank of Amsterdam 1609), and wind-powered sawmills (1594) enabling massive fleet construction. (Fiveable)

Scoring Rationale: Technological scores show Spain outperforming Portugal in Rise (3.75 vs. 3.5) and Peak (4.25 vs. 3.0) phases due to patio process innovation, Pacific navigation mastery, and mining infrastructure. However, Decline scores converge (2.75 vs. 2.0) as both empires fell behind Dutch and English innovation. Spain's inability to translate resource wealth into sustained technological leadership represents a key finding supporting the 1694 thesis.

Demographic patterns analysis

Peninsula population peaked before catastrophic decline

Pre-plague baseline populations reached approximately **8-8.5 million inhabitants** by 1590-1591, based on the Crown of Castile "Millones" service census. Regional distribution showed southern cities (Seville, Malaga) and Valencia exceeding 40,000 inhabitants each, while Barcelona held approximately 20,000 and Madrid 13,000 before its designation as capital. (PLOS)

The **plague of 1596-1602** struck catastrophically. Death toll estimates range from **600,000-700,000** (approximately 8-10% of population). Ruth MacKay's detailed study documents the path: arriving in Santander from the Netherlands, spreading south through center of Castile, reaching Madrid by 1599 and Seville by 1600. (Cambridge University Press) Eastern territories of Aragon were largely spared.

The **plague of 1646-1652** ("Great Plague of Seville") killed an additional **500,000** (approximately 5% of population). Valencia lost 30,000; Seville and districts lost 150,000 from a base of 600,000. (TheReaderWiki) Total seventeenth-century demographic impact: minimum **1.25 million plague deaths**. Population "scarcely budged" from 1596-1696. (Wikipedia)

David Reher's research identifies a critical finding: root cause was **sharp decline in marriage rates from the 1580s onward**, not just mortality. (Cairn) The urban network acted as "demographic parasite"—maintaining cities required more rural population than countryside could provide. The period 1630-1680 was worst: urban networks dragged rural population into absolute decline. (MacroTrends) Annual population growth: **0.58% (1530-1591)** collapsed to **0.09% (1591-1700)**. (ResearchGate)

Jewish expulsion removed commercially vital population

Numbers expelled remain disputed: scholarly estimates range from 40,000 (minimalist) to 200,000 (traditional), with **100,000-180,000** representing scholarly consensus. This constituted approximately **2% of total population**, but concentrated heavily in urban commercial occupations.

Contemporary chronicler Andrés Bernáldez documented Jewish occupations: "merchants, salesmen, tax gatherers, retailers, stewards of nobles, officials, tailors, shoemakers, tanners, weavers, grocers, peddlers, silk-mercers, smiths, jewelers"—notably "none tilled the earth." (EBSCO) Banking, moneylending, medicine, law, and artisanal trades concentrated in the expelled population.

Major urban centers affected included Toledo ("extinguished"), Seville, and Burgos which "lost many of their members." Capital flight was forced: four months to liquidate assets created fire-sale prices with "market saturated." The edict prohibited gold and silver export while potentially forgiving outstanding loans to the Crown—a conflict of interest noted by scholars. Ottoman Sultan Bayazet reportedly remarked: "Do they call this Ferdinand a politic prince, who can thus impoverish his own kingdom and enrich ours?"

Museum of Jewish Heritage

Joseph Pérez's revisionist assessment argues economic impact was less catastrophic than traditionally portrayed: "in view of published literature on taxation and economic activities, there is no doubt that the Jews were no longer a source of relevant [economic power]" by 1492. Yet cultural, intellectual, and professional losses were substantial, and converso populations remaining faced continued suspicion and Inquisition prosecution.

Morisco expulsion devastated Valencia while sparing Castile

The Morisco expulsion (1609-1614) removed **275,000-300,000 people**—"a much larger figure than even the Jewish expulsions in 1492." Regional impact varied dramatically.

Valencia suffered catastrophically: **33% of population expelled** (some sources cite 36%). The kingdom of Valencia transported 111,396 people in October 1609-January 1610 alone, with approximately 15,000 additional deaths from violence en route to ports. Some northern Alicante counties "lost virtually their entire population." Recovery to 1609 population levels took until approximately 1787—more than a century.

Agricultural and artisanal skills lost included sophisticated Moorish irrigation systems (Christians proved "unable to maintain the intricate irrigation systems"), silk production workforce, and intensive agriculture techniques. Long-term institutional effects documented by Harvard/PSE researchers Chaney and Hornbeck show formerly Morisco areas retained "extractive institutions" (high tax rates) after repopulation, creating "pronounced adverse selection among Christian re-settlers."

Yet **Castile** escaped significant impact—the Council of Castile concluded in 1619 that expulsion "had no economic impact" for Castile specifically. Moriscos represented only 3-4% of total Spanish population overall; concentration in Valencia created localized devastation rather than national catastrophe. (Grokikipedia)

Colonial demographic collapse fundamentally reshaped labor systems

Pre-contact population estimates remain deeply contested. The Berkeley School (Cook and Borah) proposed **25.2 million for Central Mexico in 1519** with hemispheric extrapolations "upwards of 100 million." Moderate estimates suggest 5-10 million for Central Mexico. Minimalist scholars (Rosenblat, Kroeber) proposed 4.5 million for Mexico and 8.4 million for the entire hemisphere.

Regardless of baseline, collapse was catastrophic. Cook and Borah's figures for Mexico: from 25.2 million (1518) to 700,000 (1623)—a **97% decline**. General scholarly consensus accepts 90%+ decline in many areas. [Wikipedia](#) 2019 research from University College London confirmed "some 90% of Indigenous Americans perished from wave after wave of disease," with population loss "contributing to climatic change and temporary global cooling." [Wikipedia](#)

Impact on labor systems was transformative. **Encomienda** declined as indigenous population collapsed. **Mita** (Andean forced draft labor for mines) devastated indigenous communities, creating "depopulation spirals." **African slavery** was introduced to replace collapsing indigenous labor—Cuba became a major slave destination, with Caribbean sugar production particularly dependent on African labor.

Emigration drained rather than extended population capacity

Emigration estimates: approximately **446,000** emigrants (1492-1640) [Wikipedia](#) with **240,000-250,000** in the sixteenth century alone. [Wikipedia](#) Annual rates averaged 1,000-1,500 per year in the sixteenth century, rising to approximately 3,900 per year (1601-1650).

Composition was heavily skewed: young (16-25), male, single dominated. Women never exceeded 30% of sixteenth-century emigrants. Regional origins concentrated in Andalusia (1/3 to 1/2 of all emigrants), Extremadura, and New Castile—Seville "clearly dominated the entire population flow." Later emigration shifted toward poorer northern regions (Galicia, Cantabria, Basque Country).

Emigrants were "positively self-selected" for human capital according to 2024 Economic History Review research—those choosing Mexico had higher numeracy than those choosing Peru. This suggests "brain drain" effects, though some scholars argue emigration "paled into insignificance compared to rural-urban migration."

Scoring Rationale: Demographic scores show Spain initially advantaged over Portugal (Rise 3.25 vs. 2.5) due to 8 million population base versus Portugal's 1.5 million. Scores converge through Peak (2.75 vs. 2.0) as marriage rate decline begins. Decline scores approach parity (1.75 vs. 1.5) as plague, expulsions, and stagnation compound. Spain's larger absolute numbers provided resilience Portugal lacked, but proportional losses were severe.

Information infrastructure as cross-cutting factor

The Casa de Contratación represented Spain's most sophisticated information infrastructure. Established 1503 in Seville, it maintained the **Padrón Real** (master map) from 1508, "constantly improved" and serving as "template for maps carried by every Spanish ship." Ship pilots faced 50 doblas fines for using non-official charts. The pilot examination and navigation school trained new pilots systematically, while *relaciones geográficas* surveys collected systematic geographic, linguistic, botanical, and demographic data throughout Spanish America.

Critical weakness emerged by century's end: Spanish engravers "did not have the necessary skill" to print complex maps—Diego Gutiérrez's famous 1562 "Americae Descriptio" had to be published in Antwerp.

Spanish intellectuals lamented that "maps produced by foreigners were superior to those made in Spain."

Printing under Inquisition control was less effective than traditionally portrayed. Henry Kamen argues "such strict control was impossible in practice and that there was much more liberty in this respect than is often believed." Major Golden Age authors appeared on the Index yet continued circulating. Scientific books were rarely targeted—the Index primarily aimed to protect laypeople from "misinterpreting symbolic or complex texts." However, Spain's first newspaper (*Gazeta Nueva*) appeared only in 1661, compared to Dutch newspapers from 1618.

The Dutch comparison is decisive. Amsterdam achieved information superiority through relative press freedom: "As long as news was not critical of the government and not blasphemy, it could be published." More books were printed in the Netherlands "than in all other European countries put together" by seventeenth-century estimates. Information was not restricted by "political, economic or religious authorities." Religious tolerance enabled innovation while Spanish intolerance constrained flows.

External shocks and institutional response as cross-cutting factor

Spain became history's first "serial defaulter" with royal bankruptcies in **1557, 1560, 1575, 1596, 1607, 1627, and 1647**. Evidence of institutional learning appears: the *medio general* (1577) imposed 30-58% write-offs after the 1575 bankruptcy, while the 1596 bankruptcy was milder, with average repayment of 81 cents on the dollar. Genoese *asientos* (short-term loans) and *juros* (long-term bonds) evolved as sophisticated credit instruments. Returns on sovereign lending remained "consistently profitable"—above 10% per year—explaining why banking families continued lending despite defaults.

The 1588 Armada defeat demonstrated recovery capacity: rebuilding began immediately, and "it did not take more than a year or two to rebuild" major fleets. Second (1596) and third (1597) armadas launched, though scattered by storms. Geoffrey Parker notes that "through Philip II's naval revival the English and Dutch ultimately failed to disrupt the various [Spanish] fleets of the Indies." However, approximately **10 million ducats**—roughly two years' total revenue—went to fleet rebuilding.

The **1640 cascade** revealed institutional brittleness. Catalonia (June 1640 Corpus de Sang) and Portugal (December 1, 1640) revolted simultaneously. Resources committed to Catalonia left virtually no Castilian troops present when the Portuguese revolution occurred, likely with Richelieu funding. Naples revolted in 1647, adding further strain. Each crisis compounded others: Catalonia drained troops needed for Portugal; loss of Portuguese navy weakened Atlantic defense.

Christopher Storrs's resilience thesis argues Spain was "a state and society more clearly committed to the retention of empire—and more successful in achieving this—than historians have hitherto acknowledged." Yet I.A.A. Thompson documents "debilitating shift" from crown military control to private contractors and decline of Consejo de Guerra. The 1640s demonstrated limits of Castilian financial capacity.

Session 2 scoring table

Domain	Rise (1469-1556)	Peak (1556-1598)	Decline (1598-1648)	Confidence
Social/Class	2.5	2.5	2.0	MODERATE-HIGH
Educational	3.75	4.25	3.25	MEDIUM-HIGH
Cultural	3.5	4.5	5.0	HIGH
Technological	3.75	4.25	2.75	MEDIUM-HIGH
Demographic	3.25	2.75	1.75	HIGH
Session Average	3.35	3.65	2.95	

Portuguese Comparison (Baseline from Session 1):

Domain	Rise	Peak	Decline
Social	2.5	2.5	2.0
Educational	2.0	2.0	1.5
Cultural	3.0	2.5	2.0
Technological	3.5	3.0	2.0
Demographic	2.5	2.0	1.5
Average	2.70	2.40	1.80

Spanish Advantage Over Portugal: +0.65 (Rise), +1.25 (Peak), +1.15 (Decline)

Scholarly debates engaged

Traditional versus revisionist interpretations

Hidalgo mentality debate: Traditional historiography (exemplified by Américo Castro) portrayed Spain as uniformly anti-commercial. Revisionist scholarship (Ruth MacKay, James Casey) demonstrates this was a myth created by eighteenth-century Enlightenment intellectuals, with substantial regional variation in commercial attitudes.

Inquisition impact debate: Traditional "Black Legend" accounts portrayed comprehensive suppression of intellectual life. Revisionist scholarship (Henry Kamen) demonstrates "practical limitations ensured inquisitorial censorship was in general never very effective." Recent scholarship (Cox and Figueroa 2025) synthesizes by emphasizing indirect effects through self-censorship and reduced scholarly exchange rather than direct persecution.

Colonial demographic catastrophe debate: The Berkeley School (Cook, Borah) proposed controversial high pre-contact populations (25+ million for Central Mexico; 100+ million hemispheric). Minimalist scholars (Rosenblat, Kroeber) proposed dramatically lower figures. Current scholarly consensus gravitates toward moderate estimates but accepts catastrophic 90%+ decline regardless of baseline.

Expulsion impacts debate: Joseph Pérez argues Jewish expulsion economic impact was less catastrophic than traditionally portrayed. Henri Lapeyre's work on Moriscos documents devastating regional impact in Valencia while Council of Castile (1619) concluded "no economic impact" for Castile specifically.

Golden Age paradox debate: Maravall's "cultura dirigida" thesis frames Baroque as conservative response to crisis. Critics (Melveena McKendrick) challenge the univocal paradigm, arguing for more complex relationships between cultural production and political power.

Corporate parallels for Session 2 domains

Cultural investment during decline (Golden Age paradox): IBM's research laboratory investments during market share decline in the 1990s-2000s parallel Spain's cultural flowering during economic crisis. Both represent organizations redirecting resources toward prestige projects that maintain identity and attract talent even as core business models erode.

Educational infrastructure as competitive moat: The Jesuit educational network's global reach parallels modern technology companies' developer ecosystem strategies—creating standardized training and certification systems that produce loyal, capable agents while simultaneously spreading institutional influence.

Demographic constraints on scaling: Spain's population losses from expulsions and plague parallel talent drain effects in declining technology sectors. Yahoo's and AOL's inability to retain engineering talent during decline created reinforcing cycles similar to Spain's loss of skilled Moriscos devastating Valencia's agricultural productivity.

Quantitative data tables with confidence levels

University Enrollment Data

Institution	Year	Enrollment	Confidence
Salamanca	1335	439	HIGH
Salamanca	1552	6,328	HIGH
Salamanca	1584	6,778 (peak)	HIGH
Salamanca	1601	4,000	HIGH
Salamanca	1641	3,908	HIGH
Alcalá	16th c.	Unknown	GAP
Colonial universities	16th c.	Unknown	GAP

Demographic Data

Event/Metric	Figure	Confidence
Spain population 1590	8-8.5 million	HIGH
1596-1602 plague deaths	600,000-700,000	HIGH
Jewish expulsion 1492	100,000-180,000	MEDIUM
Morisco expulsion 1609-14	275,000-300,000	MEDIUM-HIGH
Valencia population loss	33%	HIGH
Pre-contact Mexico (Cook-Borah)	25.2 million	LOW-MEDIUM
Colonial decline rate	90%+	MEDIUM-HIGH
16th c. emigration to Americas	240,000-250,000	MEDIUM

Silver Production Data

Metric	Figure	Confidence
Potosí total (1545-1810)	60,000 tons	HIGH
Peak annual output c.1600	2,700 tons	HIGH
Share of world supply (peak)	60%	HIGH
Spanish America total (1550-1800)	136,000 metric tons	HIGH
Global market share	80%	HIGH

Cultural Production Data

Figure	Output	Period
Lope de Vega	1,500-1,800 plays (400+ surviving)	1580s-1635
Calderón de la Barca	~80 autos sacramentales	1629-1681
Jesuit schools (1616)	~400 worldwide	Early 17th c.

Data gaps and intellectual honesty notes

High-priority gaps

1. **Alcalá and Valladolid enrollment:** No sixteenth-century matriculation data located despite extensive search
2. **Colonial university enrollment:** Specific student numbers for Lima, Mexico, Santo Domingo unavailable
3. **Graduate output statistics:** Number of degrees conferred annually unknown
4. **Patent/privilege records:** No systematic study of Spanish patent grants comparable to Dutch/English
5. **Pre-contact population:** Hemispheric estimates range 8 million to 112 million—fundamental uncertainty

Methodological limitations

1. **Census gap 1591-1768:** No reliable census data between these dates requires reconstruction
2. **Infant mortality underreporting:** Parish death registers systematically undercount
3. **Regional data incomplete:** Crown of Aragon data incomplete in 1591 census
4. **Converso numbers:** Proportion who converted versus emigrated remains imprecise

Scholarly controversies requiring acknowledgment

1. **Cook-Borah methodology:** Tribute record extrapolations disputed by Sanders, Zambardino
 2. **Nuptiality versus mortality:** Reher emphasizes marriage rate decline as root cause of seventeenth-century crisis—contested interpretation
 3. **Maravall's cultural thesis:** Criticized for presenting too "univocal paradigm" of Baroque as conservative project
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Implications for the 1694 thesis

The Capability Tier analysis strengthens the 1694 thesis argument. Spain possessed substantial capability advantages over Portugal: better-educated bureaucratic class, more sophisticated information infrastructure, broader cultural production, and larger demographic base. Yet these capabilities failed to produce financial innovation or sustained hegemonic position.

Key mechanism identified: Cultural and institutional resources were deployed for conservative purposes (Maravall's "cultura dirigida") rather than adaptive innovation. The Golden Age paradox reveals an empire that intensified traditional forms during crisis rather than developing new institutional capacities. Silver abundance funded consumption and warfare rather than productive investment, while serial bankruptcies demonstrated fiscal extraction without financial revolution.

The comparison to Portuguese decline confirms that Spain's additional capability resources delayed rather than prevented hegemonic transition. Both empires lacked the institutional innovations (joint-stock companies, negotiable instruments, central banking) that would characterize Dutch and later English hegemony. The Spanish case demonstrates that capability tier advantages without adaptive tier transformation produce stagnation rather than sustained power—resource abundance without institutional evolution leads to what contemporary economists call the "resource curse."

Session 3 (Adaptive Tier) will examine economic/trade systems, financial/monetary systems, and network/alliance structures to assess Spain's capacity for institutional innovation. Preliminary hypothesis: Spain's adaptive tier scores will show dramatic weakness relative to capability tier scores, explaining why substantial resources failed to produce lasting hegemonic advantage.

